Linear Algebraic Groups

Based on Lectures given by

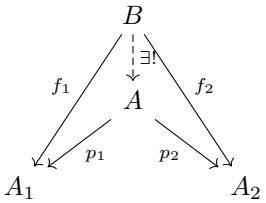
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1 Introduction

1.1 Definitions and Examples

Definition 1.1. For two objects $A_1, A_2 \in \mathscr{C}$ in a category, a product $(A, p_1 : A \to A_1, A \to A_2)$ is a triplet consists of an object and two morphisms in $\mathscr{C}$ with the following universal property. For any object $B \in \mathscr{C}$ together with morphisms $f_1 : B \to A_1, f_2 : B \to A_2$ in $\mathscr{C}$, there is a unique morphism $B \to A$ making the following diagram commutes.



By the universal property, such A is unique up to isomorphisms. We denote such object as $A = A_1 \times A_2$.

A product of n-many objects are defined recursively by the above construction. If $n = 0$, the product $1_{\mathscr{C}}$ is called a terminal object and any $A \in \mathscr{C}$, there is a unique morphism $1_A : A \to 1_{\mathscr{C}}$.

Remark 1.1. For $A \in \mathscr{C}$, $A \times 1_{\mathscr{C}} \to A$ given by the projection is an isomorphism with inverse $\mathrm{id}_A \times 1_A$.

Notation 1.1. For $A \in \mathscr{C}$, $\Delta : A \to A \times A$ is induced from $A \overset{\mathrm{id}_A}{\underset{\mathrm{id}_A}{\rightrightarrows}} A$.

Definition 1.2. Let $\mathscr{C}$ be a category admitting finite products. A group object in $\mathscr{C}$ is a quintuple (G, m, i, e) consists of an object $G \in \mathscr{C}$ and three morphisms

$$m : G \times G \to G, i : G \to G, e : 1_{\mathscr{C}} \to G,$$

satisfying the following axioms.

1. m is associative. (i.e. $m \circ (m \times \mathrm{id}_G) = m \circ (\mathrm{id}_G \times m)$).

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{\mathrm{id}_G \times m} & G \times G \\ {\scriptstyle m \times \mathrm{id}_G}\downarrow & & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow[m]{} & G \end{array}$$

2. e is a 2-sided unit. (i.e. $m \circ (\mathrm{id}_G \times e) = \mathrm{id}_G = m \circ (e \times \mathrm{id}_G)$).

$$\begin{array}{ccc} G & \xrightarrow{e \times \mathrm{id}_G} & G \times G \\ {\scriptstyle \mathrm{id}_G \times e}\downarrow & \searrow^{\mathrm{id}_G} & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow[m]{} & G \end{array}$$

3. i is a 2-sided inverse. (i.e. $m \circ (i \times \mathrm{id}_G) \circ \Delta = e \circ 1_G = m \circ (\mathrm{id}_G \times i) \circ \Delta$).

$$\begin{array}{ccccc} & & G \times G & & \\ & \overset{\Delta}{\nearrow} & & \overset{\mathrm{id}_G \times i}{\searrow} & \\ G & \xrightarrow{e \circ \mathrm{id}_G} & G & \xleftarrow{m} & G \times G \\ & \underset{\Delta}{\searrow} & & \underset{i \times \mathrm{id}_G}{\nearrow} & \\ & & G \times G & & \end{array}$$

Definition 1.3. A morphism of group objects is a morphism $f : G_1 \to G_2$ satisfying

$$m_2 \circ (f \times f) = f \circ m_1, i_2 \circ f = f \circ i_1, e_2 = f \circ e_1.$$

The subclass of objects of groups together with these morphism forms a subcategory which we will denote by $\mathrm{Grp}\mathscr{C}$.

Remark 1.2. For a group objects G in $\mathscr{C}$ and any object $A \in \mathrm{Ob}(\mathscr{C})$, $\mathscr{C}(A, G)$ inhertis a group structure as folows.

1. $e_A := e \circ 1_A, A \to 1 \to G$,
2. $f_1 \cdot f_2 := m \circ (f_1 \times f_2) \circ \Delta_A$.
3. $f^{-1} := i \circ f$.

Furthermore, the assingment $A \mapsto \mathscr{C}(A, G)$ is a contravariant functor $\mathscr{C} \to \mathrm{Grp}$ and $G \mapsto \mathscr{C}(\cdot, G)$ is a contravariant functor $\mathrm{Grp}\mathscr{C} \to \mathrm{Func}(\mathscr{C}^{\mathrm{op}}, \mathrm{Grp})$ which is an embedding (i.e. fully-faithful) by Yoneda lemma.

This is called the functor of points, we always have the abstract test group $G(1)$. In general, we will denote $G(A) := \mathscr{C}(A, G)$ for arbitrary $A \in \mathrm{Ob}(\mathscr{C})$.

Example 1.1.

Category $\mathscr{C}$	Grp$\mathscr{C}$	Terminal Object
Sets	Grp	$\{*\}$
Top	TopGrp	$\{*\}$
Man^∞	LieGrp	$\{*\}$
ComplexMfd	ComplexLieGrp	$\{*\}$
SmVar_k	AlgGrp_k	$\mathrm{Spec}(k)$
SmVar_S	AlgGrp_S	S

Remark 1.3. For Sets and $G \in \mathrm{GrpSets}$, $G(\{*\})$ contains all the information about it. That is $G(\{*\}) = \mathrm{Hom}(\{*\}, G) \cong G$.

Definition 1.4. For an algebraically closed field k, an affine algebraic k-group is a group object in the category of smooth (affine) k-varieties.

An (affine) S-group scheme is a group object in the category of Sch/S.

Definition 1.5. Let $\mathscr{C}$ be a category that admits finite co-products. A co-group object in $\mathscr{C}$ is a group object in $\mathscr{C}^{\mathrm{op}}$.

Example 1.2. Let $\mathbb{R}$ be a commutative unital ring and $\mathscr{C}$ the category of R-algebras which admits finite co-product. Then a co-group object in $\mathscr{C}$ is an R-Hopf algebra. Explicitly, for A an R-algebra, we have

$$\delta : A \to A \underset{R}{\otimes} A, \varepsilon : A \to R, \eta : A \to A$$

which are called comultiplication, counit, and coinversion (or antipodal), respectively.

Remark 1.4. Recall that we have an equivalence of categories,

$$(R\text{−Alg})^{\mathrm{op}} \overset{\mathrm{Spec}}{\to} \mathrm{AffSch}/R$$

inducing an equivalence between R-Hopf algebra and affine R-group schemes.

Remark 1.5. Let G be an affine k-variety and $\mathcal{O}_G$ be its ring of functions. Let $f \in \mathcal{O}_G, f : G \to k$, then

$$f \circ m : G \times G \to k \in \mathcal{O}_{G\times G} \cong \mathcal{O}_G \underset{k}{\otimes} \mathcal{O}_G.$$

Example 1.3. Let R be a commutative unital ring, $\underline{\mathbb{G}_m}$ be the functor such that

$$R\text{−Alg} \ni S \overset{\mathbb{G}_m}{\mapsto} (S^\times, \cdot) \in \mathrm{Grp}$$

is the functor of points of the R-group schemes with Hopf-algebra,

$$A := R[x, x^{-1}].$$

Structure morphisms are given by

$$A \ni x \overset{\delta}{\mapsto} x \otimes x \in A \otimes A, A \ni x \overset{\varepsilon}{\mapsto} 1_R \in R, x \overset{\eta}{\mapsto} x^{-1} \in A.$$

Example 1.4. Let R be a commutative unital ring, $\underline{\mathbb{G}_a}$ be the functor such that

$$R\text{−Alg} \ni S \overset{\mathbb{G}_m}{\mapsto} (S, +) \in \text{Grp}$$

is the functor of points of the affine R-group schemes with Hopf-algebra,

$$A := R[x].$$

Structure morphisms are given by

$$A \ni x \overset{\delta}{\mapsto} x \otimes 1_A + 1_A \otimes x \in A \otimes A, A \ni x \overset{\varepsilon}{\mapsto} 0_R \in R, x \overset{\eta}{\mapsto} -x \in A.$$

Example 1.5. Let R be a commutative unital ring, $\underline{\mathrm{GL}_N}$ be the functor such that

$$R\text{−Alg} \ni S \overset{\mathrm{GL}_N}{\mapsto} \mathrm{GL}_N(S) \in \text{Grp}$$

is the functor of points of the affine R-group schemes with Hopf algebra,

$$A = R[x_{11}, \cdots, x_{NN}, \det(x_{ij})^{-1}].$$

The structure morphisms are,

$$\begin{aligned}
\delta(x_{ij}) &= \left(\sum_{k=1}^{N} x_{ik} x_{kj} \right), \\
\varepsilon(x_{ij}) &= (\delta_{ij}), \\
\eta(x_{ij}) &= (\det(x_{ij}))^{-1} \mathrm{adj}(x_{ij}).
\end{aligned}$$

Example 1.6. $\underline{\mathrm{SL}_N}$ is the functor sending

$$R\text{−Alg} \ni S \overset{\mathrm{SL}_N}{\mapsto} \mathrm{SL}_N(S) \in \text{Grp}.$$

Its Hopf algebra is

$$A = R[x_{ij}]_{ij}/(\det(x_{ij}) - 1).$$

Example 1.7. $\underline{\mu_N}$ is the functor sending

$$R\text{−Alg} \ni S \overset{\mu_N}{\mapsto} \{x \in S^\times \mid x^N = 1\} \in \text{Grp}.$$

Its Hopf algebra is

$$A = R[x]/(x^N - 1)$$

and the structure morphisms are the same as in $\underline{\mathbb{G}_m}$.

Example 1.8. $\underline{\alpha_p}$ for prime p of the characteristic of R is the functor sending

$$R\text{−Alg} \ni S \overset{\alpha_N}{\mapsto} \{x \in S^\times \mid x^p = 0\} \in \text{Grp}.$$

Its Hopf algebra is

$$A = R[x]/(x^p)$$

and the structure morphisms are the same as in $\underline{\mathbb{G}_a}$.

1.2 Basic Geometric Properties of Group Schemes

Definition 1.6. A scheme is separated if the diagonal map $\Delta : X \to X \times X$ is a closed immersion.

Remark 1.6. A topological space is Hausdorff if and only if $\Delta : X \to X \times X$ is closed.

Lemma 1.1. An S-group scheme G is separated if and only if $e : S \to G$ is a closed immertion. This is automatic if $S = \mathrm{Spec}(k)$ or G is affine.

Proof.

$$\begin{array}{ccc} G & \longrightarrow & S \\ {\scriptstyle\Delta}\downarrow & & \downarrow{\scriptstyle e} \\ G \underset{S}{\times} G & \xrightarrow[m\circ(\mathrm{id}_G, i)]{} & G \end{array} \qquad \begin{array}{ccc} S & \xrightarrow{e} & G \\ {\scriptstyle e}\downarrow & & \downarrow{\scriptstyle\Delta} \\ G & \xrightarrow[(\mathrm{id}_G \times (e\circ 1_G))]{} & G \underset{S}{\times} G \end{array}$$

Since closed immersion is stable under base change, the equivalence is done. □

Remark 1.7. A topological gropu$G$ is Hausdorff if and only if $e \in G$ is a closed point (i.e. $\{e\}$ is closed).

Recall that X scheme is connected or irreducible if $|X|$ has that property, where $|X|$ is the underlying space. In general, irreducibility implies the connectedness.

Lemma 1.2. For a S-group scheme, a connected component is an irreducible component. In particular, these two notions are equivalent.

Proof. Since an irreducible component is not fully contained in the union of two others, there is a point contained only on an irreducible component. By translation every point lies in a unique irreducible component, thus the irreducible components are disjoint. □

Remark 1.8. If G is smooth affine, irreducibility, connectedness and $\mathcal{O}_G$ being an integral domain are all equivalent.

Identity component Let X be a scheme and $Z \subseteq |X|$ be a closed subset, let $Z \hookrightarrow X$ be a canonical inclusion as topological spaces. Then $(Z, i^*\mathcal{O}_X)$ is a locally ringed space but may not be a scheme. To make this a scheme we need $(Z, i^*(\mathcal{O}_X/\mathcal{I}_Z))$ where

$$\mathcal{I}_Z(U) = \{f \in \mathcal{O}_X(U) \mid \forall x \in U \cap Z, f(x) = 0 \text{ in } \mathcal{O}_{X,x}/\mathfrak{m}_x\}.$$

This is called the reduced scheme structure on Z, however, if Z is generalizing (closed under generalization) then $(Z, i^*\mathcal{O}_X)$ is a scheme.

Lemma 1.3. A connected component is generalizing.

Proof. Since Z and $\overline{\{y\}}$ are both connected and meet at x, $Z \cup \overline{\{y\}}$ is also connected hence $Z = Z \cup \overline{\{y\}}$ thus $y \in Z$. □

Definition 1.7. For a k-group scheme G, the identity component G^0 is defined as the scheme $(|G|^0, \mathcal{O}_G|_{|G|^0})$ which is a group scheme.

Recall if $(R, \mathfrak{m})$ is a local ring, then the dimension of R is the maximal chain of prime ideals which is called the Krull-dimension. Also $\mathfrak{m}/\mathfrak{m}^2$ is a k-vector space where $k = R/\mathfrak{m}$, hence we can consider the dimension of $\mathfrak{m}/\mathfrak{m}^2$ as a k-vectorspace. R is said to be regular if these two dimensions coincide.

Definition 1.8. A scheme X is regular if all local rings $\mathcal{O}_{X,x}$ are regular.

Definition 1.9. A k-scheme X of finite type is smooth if $X \underset{k}{\times} \overline{k}$ is regular.

Definition 1.10. For S-scheme X, geometric fibers are the base change $X_{\overline{S}}$

$$\begin{array}{ccc} X_{\overline{S}} & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & S \end{array}$$

where k is algebraically closed.

Definition 1.11. A S-scheme X is smooth if it is flat, locally finite presentation and its geometric fibers are regular.

Lemma 1.4. For the notion of smoothness for group scheme, the geometric fibers being regular can be replaced by the following criteria. Every point $\overline{e}$ of a geometric base change $X_{\overline{S}}$ over e, $X_{\overline{S}}$ is regular at $\overline{e}$.

Proof. Clearly if G is smooth, the above condition is met. Conversely, for any $g \in G(k)$, where k is algebraically closed, we have

$$L_g : G \underset{S}{\times} \mathrm{Spec}(k) \xrightarrow{\sim} G \underset{S}{\times} \mathrm{Spec}(k).$$

Therefore, $\mathcal{O}_{G,g}$ is regular. The smooth locus $U \subseteq G$ is open, non-empty. Take $V = G \backslash U$. If V is not empty then some affine open has a k-point for some algebraically closed field k by Nullstellensatz hence a contradiction. □

Definition 1.12. A scheme X is reduced if all its stalks are reduced as rings. Equivalently all affine opens are spectrum of reduced rings.

Remark 1.9. Even though X is reduced, for a morphism of schemes $X \to Y$, the fibers might not be reduced. The following is a counter example,

$$(\mathbb{Z}[x]/(x^p - 1)) \otimes \mathbb{F}_p.$$

Remark 1.10. X/k is reduced but $X \underset{k}{\times} \overline{k}$ needs not be reduced. We can construct a counter example as follows.

$$k = \mathbb{F}_p(s), R = k[t]/(t^p - s)$$

is reduced however,

$$R \underset{k}{\otimes} k(\sqrt[p]{s})$$

is not.

For a scheme X over k, geometrically reduced implies reduced. If k is perfect then these two notions coincide.

Lemma 1.5. A k-group scheme of locally finite type is smooth if and only if it is geometrically reduced.

Proof. By the definition of smoothness, we assume k to be algebraically closed without loss of generality. Recall that regular local ring is reduced. Conversely, a smooth scheme X is reduced which is locally of finite type, then it has non-empty smooth locus which is open and dense. If $X = G$ a group scheme then using translation, all points are smooth. □

Proposition 1.1. For a k-group scheme G locally of finite type where k is algebraically closed, G satisfies the following properties.

1. G is equidimensional and $\dim(G) \leq \dim T_e(G)$.
2. The following statements are equivalent for G
   i. G is smooth.
   ii. G^0 is smooth.
   iii. $\mathcal{O}_{G,e}$ is regular.
   iv. $\dim(G) = \dim T_e(G)$.
   v. G is reduced.
   vi. For any k-algebra R and $I \subseteq R$ an ideal such that $I^2 = 0$, $G(R) \twoheadrightarrow G(R/I)$.

Proof. G is equi-dimensional by translation. By Nakayama's lemma $\dim T_e(G) \geq \dim(\mathcal{O}_{G,e})$.

For i $\iff$ ii $\iff$ iii are due to translation. iii $\iff$ iv is by definition. i $\iff$ v is by Lemma 1.3. i $\iff$ vi by lifting criterion. □

Let R be a finite k-algebra and $X = \mathrm{Spec}(R)$ where k is algebraically closed. Since $x \in X_{\mathrm{cl}}$ corersponds to a maximal ideal in R, the composition,

$$k \to R \to R/\mathfrak{m}_x$$

is an isomorphism. We write $f(x) = \mathrm{ev}_x(f) \in k$.

Definition 1.13. A derivation $d : R \to k$ is a k-linear map satisfying Leibniz rule.

In the above setting, we have

$$T_x X = \mathrm{Der}_k(R, k)$$

and restriction to $\mathfrak{m}_x$ gives an isomorphism of $\mathrm{Der}_k(R,k)$ and $\mathrm{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k)$.

A vector field is $d : x \mapsto d_x$ such tha for each $f \in R$, $d(f) \in R$. We have a morphism,

$$\mathrm{Der}_k(R,R) \overset{d \mapsto \mathrm{ev}_x \circ d}{\to} \mathrm{Der}_k(R,k).$$

Proposition 1.2 (Cartier)**.** Let k be a field of characteristic 0. Any (affine) k-group scheme of finite type is automatically smooth.

Proof. As usual, we assume k is algebraically closed. Let $I \subseteq \mathcal{O}_G$ be nilradical, then $I \subseteq \mathfrak{m}_{G,e} \subseteq \mathcal{O}_G$.

Claim 1. $I \subseteq \mathfrak{m}_{G,e}^2$

Proof: Let $d_e \in T_eG = \mathrm{Der}_k(\mathcal{O}_G, k) = \mathrm{Hom}_k(\mathfrak{m}_{G,e}/\mathfrak{m}_{G,e}, k)$. Hence enough to show that $d_e(I) = 0$. Take $d \in \mathrm{Der}_k(\mathcal{O}_{G,e}, \mathcal{O}_{G,e})$ that lifts d_e, using Leibniz rule we see $d_e(I) = 0$. Indeed for $f \in I$, if N is the smallest natural number such that $f^N = 0$, then

$$d_e(f^N) = Nf^{N-1}d(f),$$

Note that N is not 0 as k is of characteristic 0 and f^{N-1} is not 0. Hence $d(f)$ is not a unit. Hence $d(f) \in \mathfrak{m}_{G,e}$. Recall we have $d_e(f) = \mathrm{ev}_e(d(f)) = 0$. ■ Let $H = G_{\mathrm{red}}$ which is a reduced k-group scheme hence smooth by Lemma We have $\mathcal{O}_H = \mathcal{O}_G/I$, hence $\mathfrak{m}_{H,e} = \mathfrak{m}_{G,e}/I$. With the claim we have $\mathfrak{m}_{H,e}^2 = \mathfrak{m}_{G,e}^2/I$. Therefore $\mathfrak{m}_{H,e}/\mathfrak{m}_{H,e}^2 \cong \mathfrak{m}_{G,e}/\mathfrak{m}_{G,e}^2$. Now $\dim G \leq \dim T_e(G) = \dim T_e(H) = \dim(H) = \dim G$ by Proposition 1.1. Hence $\dim G = \dim T_e(G)$ hence G is smooth. □

1.3 Kernels and Exact Sequences

Definition 1.14. The kernel of a homomorphism of algebraic group schemes $\varphi : G \to H$ is the fiber product

$$\begin{array}{ccc} \ker\varphi := G \underset{H}{\times} \{*\} & \longrightarrow & \{*\} \\ \downarrow & & \downarrow \\ G & \xrightarrow[\varphi]{} & H \end{array}$$

Remark 1.11. Obviously $\ker\varphi(R) = \ker(\varphi(R) : G(R) \to H(R)) \trianglelefteq G(R)$ and hence $\ker\varphi \trianglelefteq G$.

Remark 1.12. If G, H are both affine then so is $\ker\varphi$ with the underlying Hopf algebra

$$\mathcal{O}_{\ker\varphi} = \mathcal{O}(G) \underset{\mathcal{O}(H)}{\otimes} k \cong \mathcal{O}(G)/I_H\mathcal{O}(G),$$

where $I_H := \ker(\mathcal{O}(H) \to k)$ which is called the augumentation ideal, (i.e., the kernel of morphism $\mathcal{O}(H) \ni f \mapsto f(e) \in k$).

Example 1.9. $\mu_n = \ker(\mathbb{G}_m \overset{(\cdot)^n}{\to} \mathbb{G}_m)$ and $\alpha_p = \ker(\mathbb{G}_A \overset{(\cdot)^p}{\to} \mathbb{G}_A)$ where the latter group scheme is over a field of characteristic p.

Definition 1.15. A morphism of schemes is faithfully-flat if it is flat and surjective as a morphism between underlying topological spaces.

Definition 1.16. The sequence

$$e \to N \xrightarrow{i} G \xrightarrow{q} Q$$

is exact if i induces an isomorphism between N and $\ker q$.

The sequence

$$e \to N \xrightarrow{i} G \xrightarrow{q} Q \to e$$

is exact if in addition q is faithfully-flat.

Proposition 1.3. Given an exacr sequence,

$$e \to N \to G \to Q \to e,$$

1. N, Q are smooth $\Rightarrow$ G is smooth.
2. G is smooth $\Rightarrow$ Q is smooth.

Proof. **1.** If q is flat and fibers of q are the translates of N hence smooth. Note that q is smooth if N is smooth and hence G is smooth as Q is.

2. If q is faithfully flat then $\mathcal{O}_Q \hookrightarrow q_*\mathcal{O}_G$ is injective. As G is smooth, it is geometrically reduced, so is $\mathcal{O}_Q$ hence Q is smooth. □

Example 1.10. An example of G being smooth but N fails to be is $\mathbb{G}_m$ and μ_p where the involving field is of characteristic p.

Proposition 1.4. Let $\varphi : G \to H$ be a surjective morphism of group varieties. The following statements about φ are equivalent.

1. $d_e\varphi : T_eG \to T_eH$ is surjective.
2. $\ker \varphi$ is smooth.
3. φ is smooth.

Proof. WIthout loss of generality, we assume k is algebraically closd. Write $N = \ker \varphi$. Recall the equality agian that the fibers of φ are the translates of N hence all dimensions of fibers are equal to $\dim(N)$. In particular, we obtain,

$$\dim(G) = \dim(H) + \dim(N).$$

Recall by Proposition 1.1

$$\dim T_eG \geq \dim G.$$

The equality holds if and only if G is smooth. Consider

$$\begin{array}{ccccccc}
0 & \longrightarrow & T_eN & \longrightarrow & T_eG & \xrightarrow{d_e\varphi} & T_eH \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & N(k(\varepsilon)) & \longrightarrow & G(k(\varepsilon)) & \longrightarrow & H(k(\varepsilon)) \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & N(k) & \longrightarrow & G(k) & \longrightarrow & H(k)
\end{array}$$

We have $\dim T_eG \leq \dim T_eN + \dim T_eH$ and the equality holds if and only if φ is surjective. Using once again Proposition 1.1, we see

$$\dim G \leq \dim T_eN + \dim H,$$

hence **i** $\iff$ **ii**.

ii $\iff$ **iii**. φ smooth means fibers are smooth hence $\Leftarrow$. Conversely, $d_e\varphi$ being surjective means that $d_g\varphi$ is surjective for all $g \in G(k)$ by the homogeneity, and as G, H are smooth so is φ. □

1.4 Group Actions

Definition 1.17. An action of a k-group scheme G on a k-scheme X is a regular map $\mu : G \times X \to X$ that the following diagrams are commutative.

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{\mathrm{id}_G \times \mu} & G \times X \\ {\scriptstyle m\times \mathrm{id}_X}\downarrow & & \downarrow{\scriptstyle \mu} \\ G \times X & \xrightarrow[\mu]{} & X \end{array} \qquad \begin{array}{ccc} \{*\} \times X & \xrightarrow{e\times \mathrm{id}_X} & G \times X \\ & \searrow^{\cong} & \downarrow{\scriptstyle \mu} \\ & & X \end{array}$$

We will write $g\cdot x, gx, g\times x$ for $\mu(g, x)$. Giving group scheme actions is equivalent to giving actions $G(R) \circlearrowright X(R)$.

Lemma 1.6. Consider $G \overset{\mu}{\circlearrowright} X$ which is faithfully flat. Then we have the following statements about μ.

1. G is smooth then μ is smooth.
2. G is finite then μ is finite.

Proof. If $(g, x) \mapsto (g, g \cdot x)$ is an isomorphism then by the diagram

$$\begin{array}{ccc} G \times X & \xrightarrow{(g,x)\mapsto(g,g\cdot x)} & G \times X \\ {\scriptstyle \mu}\downarrow & & \downarrow{\scriptstyle p_X} \\ X & \xrightarrow[\mathrm{id}_X]{} & X \end{array}$$

any properties of μ hold for p_2 and therefore for G as well. □

Definition 1.18. A morphism of k-schemes is G-equivariant if it commutes with G-actions.

Proposition 1.5. $X \xrightarrow{f} Y$ be G-equivariant between non-empty k-schemes.

1. Y is reduced and $G(\overline{k}) \circlearrowright Y(\overline{k})$ is transitive then f is faithfully flat.
2. If $G(\overline{k}) \circlearrowright X(\overline{k})$ is transitive then $f(|X|) \subseteq |Y|$ locally closed and hence $f(X)_{\mathrm{red}} := f(|X|)$ with its reduced scheme structure.

3. If $G(\overline{k}) \circlearrowright X(\overline{k})$ is transitive and X is reduced then f factors as

$$X \overset{\text{faithfully flat}}{\to} f(X)_{\text{red}} \hookrightarrow Y,$$

and $f(X)_{\text{red}}$ is G-stable.

Proof. See [2]. □

Definition 1.19. Let G be a group scheme and X be a scheme where G acts on. For $x \in X(k)$, we have the orbit map is

$$G \ni g \overset{\mu_x}{\mapsto} gx \in X.$$

μ_x is the restriction of μ to $G \times \{x\}$. If $G(\overline{k})$ acts transitively on $X(\overline{k})$ then μ_x is surjective.

The orbit $G \cdot x$ is the image of $|\mu_x|$.

Remark 1.13. Due to Proposition 1.5.2, $|\mu_x|$ is locally closed together with considering the trivial action of G on itself.

Example 1.11.

$$\mathrm{SL}_2 \times \mathbb{A} \ni \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

Then we have two orbits,

$$\{0\}, \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \neq 0 \right\}.$$

The first one is clearly closed however, the latter has the closure the whole hence not closed even it is locally closed.

Proposition 1.6. Let G be a k-group variety acting on k-variety X where k is algebraically closed. Then every orbit of minimal dimension is closed. In particular, there will always be a closed orbit.

Proof. Let $O := G \cdot x$ be an orbit. Then there is $U \subseteq \overline{O}$ a dense open subset contained in O by Chevalley's theorem. Observe that

$$O = \bigcup_{g \in G(k)} gU \subseteq \overline{O},$$

hence open. Therefore $\overline{O} \backslash O$ is closed and is of dimension less than $\dim O$. If O is closed then there is nothing to show. Otherwise $\overline{O} \backslash O \neq \emptyset$ is a union of orbit of dimension less than $\dim O$ hence O is closed if it is of minimal dimension. □

Theorem 1.1 (Homomorphism Theorem)**.** Let $G \overset{\varphi}{\to} H$ be a homomorphism of group schemes. if G is smooth then φ factors as,

$$G \overset{\text{faithfully flat}}{\to} I \overset{\text{closed immersion}}{\to} H.$$

References

[1] Linear algebraic groups.

[2] Linear algebraic groups.